

FinGuard (FraudShield-SEBI Pro) — Grand Finale Evaluation & Roadmap Report

1. Grand Finale Evaluation Scorecard

The MIT Delhi Grand Finale evaluates 30 shortlisted teams across 2 leaner phases to measure **depth of execution and product-grade readiness** rather than a fresh concept. The evaluation matrix below breaks down the four core scoring pillars, FinGuard's current standing, and the target score needed for a Top 3/Top 5 podium finish.

Evaluation Dimension	Weight / Max Marks	Current Estimated Score	Target Score (Top 3)	Critical Gap
1. Problem Statement Completeness	35	24	34	Missing the "Fake Trading App & APK Indicator" pillar explicitly mandated in Problem Statement #7.
2. Technical Rigor & Edge Efficiency	30	21	28	Local multi-model VRAM contention and latency bottlenecks across 15 frames on commodity hardware.
3. Regulatory Grounding & Data Integrity	20	17	19	Static offline JSON database lacks a realistic delta synchronization mechanism for active SEBI registrations.
4. Live Showcase & Impact Narrative	15	10	14	Over-focusing on back-office B2B compliance while missing the visible, emotional citizen-protection narrative.
Total Score	100	72 / 100	95 / 100	<i>Current: Top 15 Qualifier → Target: Grand Finale Winner</i>

2. Gap Analysis: Where We Currently Lack

Pillar 1: Incomplete Problem Statement Coverage (-11 Marks)

- Problem Statement #7 explicitly demands three detection vectors: (a) Deepfake media, (b) Impersonation text patterns, and (c) **Fake trading-app indicators**.
- FinGuard handles video, audio, and text exceptionally well, but completely ignores fake broker APKs, domain typo-squatting, and malicious trading web apps. Missing an entire clause of the official prompt guarantees point deductions during the initial rubric review.

Pillar 2: Latency & VRAM Contention (-9 Marks)

- Running MTCNN, Dual-Stream Vision (EfficientNet + Xception), Wav2Vec2, faster-whisper, and LLaMA 3.1 sequentially creates severe latency on local GPUs.
- Processing a 30-second clip across 15 extracted frames live on stage risks exceeding the 2-minute mark, exposing the product to skepticism from enterprise tech judges about high-throughput readiness.

Pillar 3: Static Regulatory Data vs. Live Reality (-3 Marks)

- Relying exclusively on static_data/sebi_registry.json creates a vulnerability during Q&A. SEBI adds, suspends, and revokes Research Analyst (RA) and Investment Adviser (IA) licenses weekly. Judges will ask how an "air-gapped" system maintains accuracy without stale records.

Pillar 4: B2B Coldness vs. Social Cause Emotional Resonance (-5 Marks)

- Pure corporate SaaS pitches often lose subjective scoring to emotionally compelling social-cause projects (e.g., accessibility tools).
- Presenting FinGuard strictly as an affiliate marketing filter masks its primary identity: an investor-protection shield saving middle-class life savings from financial ruin.

3. Step-by-Step Refinement Roadmap (The Path to 95+)

You do not need to rewrite the architecture. The existing 8-stage pipeline is strong. The roadmap below focuses on bolting on the missing problem statement pillar, optimizing latency, and executing a dual-impact demo.

[Week 1-2: Feature Completion] → [Week 3: Latency Optimization] → [Week 4: Data Sync] → [Week 5-6: Showcase Polish]

- Fake APK/URL Scanner (Stage 9)
- Early-Exit Vision Ingestion
- SEBI Delta Scraper
- Retail Bot + Enterprise UI
- Domain Typo-Squatting Engine
- VRAM Model Resident Caching
- Cron Sync Schema
- 5-Min Dual-Impact Script

Weeks 1–2: Complete the Problem Statement Trinity (Stage 9: App & Domain Inspector)

- **APK Metadata Parser:** Add a lightweight backend module using androguard or apkutils to inspect uploaded Android .apk packages for deceptive package names (e.g., com.zerodha.vipinvestment vs. com.zerodha.kite3) and dangerous background permissions (reading SMS/Contacts).
- **Domain Typo-Squatting Detector:** Use Levenshtein distance against a whitelist of

SEBI-registered broker domains to instantly flag phishing URLs extracted via OCR (e.g., groww-bonus-login.xyz vs. groww.in).

- **Result:** Achieves 100% formal compliance with all three prongs of Problem Statement #7.

Week 3: Latency Optimization & Early-Exit Execution

- **Adaptive Frame Exit:** Implement an early-exit threshold in stage2_vision.py. If the first 4 keyframes return a deepfake probability >0.92, terminate further frame inference and finalize the score.
- **Pre-Loaded Model State:** Ensure all PyTorch and ONNX models are loaded into VRAM on FastAPI startup rather than initialized per-request to keep end-to-end demo latency under 10 seconds.

Week 4: Dynamic Registry Synchronization

- **SEBI Sync Script (scripts/sync_sebi_registry.py):** Build an automated Python updater that pulls official intermediary rosters from SEBI's public portal and writes incremental deltas into static_data/sebi_registry.json.
- **Defense Strategy:** Defend the architecture as *"Air-gapped execution with an authenticated one-way daily synchronization daemon."*

Week 5: The "Dual-Impact" Interface (Citizen Tipline + B2B Gateway)

- **Retail Tipline Bot:** Deploy a lightweight Telegram/WhatsApp bot webhook connected to app.py.
- **User Flow:** A user forwards a suspicious reel or APK link \rightarrow The bot returns an instant forensic verdict in Hinglish + a 1-click button to download the pre-formatted SEBI SCORES complaint draft.

Week 6: Grand Finale Polish & Rehearsal

- **Stress-Test Folder Suite:** Run scripts/run_folder_test.py across 50+ compressed WhatsApp clips to lock in false-positive thresholds for the vision/audio consensus matrix.
- **Pitch Alignment:** Structure the 5-minute presentation around the social crisis of retail investor fraud, live end-to-end interception, and full compliance with SEBI mandates and the DPDP Act 2023.

4. Summary Deliverables for Delhi

- **Stage 9 Module:** APK and Phishing Domain Scanner integrated into the main pipeline.
- **Sub-10s Latency:** Early-exit vision scoring and pre-warmed GPU memory.
- **Dual Demo UI:** The B2B Enterprise Audit Dashboard alongside an active Telegram Retail Tipline Bot.
- **Legal Shield:** 1-Click SEBI SCORES compliant legal drafts and automated DPDP PII redaction.